UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE O Level

MARK SCHEME for the November 2005 question paper

5070 CHEMISTRY

5070/04 **Paper 4 maximum raw mark 60**

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1 **(a)** 46 (1) cm^3

(b) less (1) rate decreases as reaction proceeds (1) or similar.

(c) **(i)** 0.005 (1)
(ii) 100 (1)
(iii) 120 (1) cm^3

(d) **(i)** more powdered (1)
(ii) increase concentration (1) **[8]**

2 **(a)** **(i)** hydrogen (1)
(ii) pops in flame (1)
(iii) magnesium (1)
(iv) Ag/Pb (1) reference to Reactivity series (1)

(b) **(i)** III/IV/V (1)
(ii) Zn (1), reason based on relative reactivities (1)
(iii) displacement or redox (1)
(iv) Produces zinc oxide and carbon dioxide (1)
$ZnCO_3 \rightarrow ZnO + CO_2$ (1)

(c) **(i)** carbon monoxide or dioxide(1)
(ii) burns with a blue flame or lime water turns milky (1)
(iii) $Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$ (2) **or**
$2Fe_2O_3 + 3C \rightarrow 4Fe + 3CO_2$ (2) **[15]**

3 to 6 **(c), (b), (b), (c).** [1 mark for each] **[4]**

7 **(a)** 2.05g (1)

(b) yellow to orange, red or pink (1)

(c)

25.8	47.0	32.3
0.0	21.8	6.9
25.8	25.2	25.4

[1 mark for each correct row or column] (3)

Mean value 25.3 (1) cm^3

(d) 0.0024 (1)

(e) 0.0012 (1)

(f) 0.012 (1)

(g) 170.8 (1)

(h) **(i)** 137 (1),
(ii) Barium (1) **[12]**

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8 1 coloured (1) solution, effervescence (1)
test: lime water, turns milky (1) carbon dioxide (1)

2 green precipitate (1) insoluble in excess (1)

3 green precipitate (1) insoluble in excess (1)

$FeCO_3$ (1) **[9]**

9 **(a)** 27.8, 30.6, 33.3, 34.0 [all correct] (1)
Temp rises: 2.8, 5.6, 8.3, 9.0, 9.0 [all correct] (1)

(b) points correctly plotted (1)
two straight lines intersecting correctly (2)

(c) **(i)** 0.29 (1)g
(ii) 0.65 (1)g
(iii) reaction complete
or all copper(II) sulphate reacted (1)

(d) zinc dissolves, reacts, disappears
solution becomes less blue to colourless,
copper, or red deposit or solid collects on floor of beaker; [any 2] (2)

(e) 0.56 (1)g which is 0.01 moles or similar explanation based on **(c)(ii)** (1) **[12]**

[For answers **(c)(i)** and **(ii)** please read candidate's graph to nearest half square.]

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